



DEPARTMENT OF PHYSICS

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Master's Thesis

Simulation of Radiative Effects for the AMBER Proton Radius Measurement with McMule

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Abstract

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1 Introduction

The proton is one of the basic building blocks of ordinary matter and, at the same time, a highly complex system. It is a composite particle made of quarks and gluons, with a valence-quark content of two up quarks and one down quark, bound together by the strong interaction. While its constituents and their interactions are described by quantum chromodynamics (QCD), many properties of the proton cannot be obtained straightforwardly from perturbative calculations. Precise measurements of its electromagnetic properties therefore provide valuable information about its structure and are essential for testing our theoretical understanding of the proton.

Since the electric charge of the proton is distributed over a finite volume rather than concentrated at a single point, its spatial extent can be characterized by the proton charge radius. The electromagnetic structure of the proton is described by form factors, which parameterize its charge and magnetization distributions. In particular, the electric Sachs form factor $G_E(Q^2)$ describes the electric structure of the proton as a function of the squared four-momentum transfer Q^2 . Its slope at zero momentum transfer is directly related to the proton charge radius,

$$r_p^2 = -6 \left. \frac{dG_E(Q^2)}{dQ^2} \right|_{Q^2=0}.$$

A precise determination of the proton charge radius therefore requires measurements at very small momentum transfers and an accurate extrapolation of the electric form factor towards $Q^2 = 0$.

Experimentally, the proton charge radius can be determined using two different approaches. The first is based on spectroscopy, in which atomic energy differences are determined by measuring the absorption or emission of electromagnetic radiation. A particularly sensitive observable is the Lamb shift. In the Dirac description of hydrogen with a point-like proton, the $2S_{1/2}$ and $2P_{1/2}$ states are degenerate. However, in 1947, Lamb and Retherford experimentally demonstrated that the $2S_{1/2}$ state lies slightly above the $2P_{1/2}$ state in energy. This observation played an important role in the development of quantum electrodynamics (QED), which explains most of this energy difference through radiative corrections.

The proton charge radius introduces an additional contribution to these atomic energy levels. This effect is especially important for S -states, whose wave functions have a non-zero probability density at the position of the proton and therefore overlap with its charge distribution. The $2P$ -state, in contrast, has a vanishing probability density at the origin and is affected much less. As a result, the measured energy difference between the $2S$ and $2P$ states also contains information about the proton charge radius. By comparing precisely measured transition frequencies with theoretical predictions, the proton radius can therefore be extracted.

Measurements of the Lamb shift in muonic hydrogen provide an especially precise determination of the proton radius. Muonic hydrogen is a hydrogen-like atom in which the electron is replaced by a negatively charged muon. Due to the much larger mass of the muon compared to the electron, its Bohr radius is approximately 200 times smaller than that of ordinary hydrogen. As a result, the muon has a significantly larger probability density close to the proton, making the energy levels much more sensitive to the finite proton size.

The second approach uses elastic lepton–proton scattering experiments, where the differential scattering cross section provides access to the proton’s electromagnetic form factors. More details on this approach will be discussed in Chapter 3.

Historically, measurements based on elastic electron–proton scattering and spectroscopy of ordinary hydrogen yielded a proton charge radius of approximately 0.88 fm. This picture changed significantly in 2010, when a high-precision measurement of the Lamb shift in muonic hydrogen led to a significantly smaller value of approximately 0.84 fm. The discrepancy between this result and previous determinations became known as the proton radius puzzle. Figure 1.1 illustrates this discrepancy and summarizes several experimental determinations of the proton charge radius.

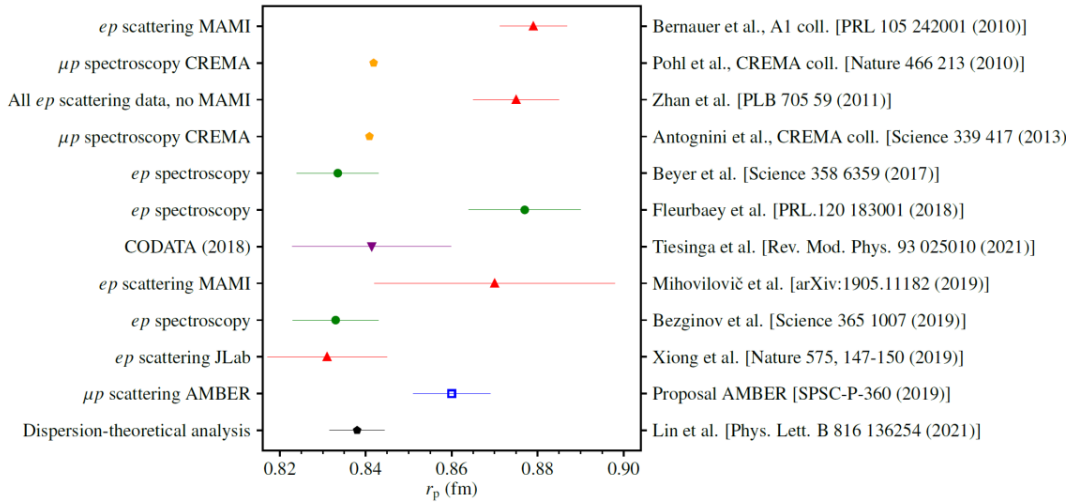


Figure 1.1: Comparison of proton charge radius measurements from different experiments. Figure taken from [Karl PAW25].

Although more recent measurements have substantially reduced the discrepancy, an independent high-precision determination of the proton radius remains important. In this context, elastic muon–proton scattering provides such an independent approach. A new precision measurement using this method is pursued by the AMBER experiment at CERN, which will be presented in detail in Chapter 2.

2 The AMBER Experiment

2.1 Overview of the AMBER Experiment

The AMBER (Apparatus for Meson and Baryon Experimental Research) experiment, designated NA66, is a fixed-target experiment located in the EHN2 experimental hall in the North Area of CERN, as shown in Figure 2.1. It uses the M2 beam line of the Super Proton Synchrotron (SPS), which provides high-energy muon and hadron beams to the experiment.

AMBER builds on the infrastructure and large-acceptance spectrometer of the former COMPASS experiment and extends it with detector systems developed for its new physics program. Its measurements cover several topics in hadron physics, including hadron structure, spectroscopy, and a high-precision determination of the proton charge radius. For the proton-radius measurement, AMBER studies muon–proton scattering at very small four-momentum transfers $1 \times 10^{-3} \text{ GeV}^2 < Q^2 < 4 \times 10^{-2} \text{ GeV}^2$, where the electric form factor is particularly sensitive to the proton charge radius.

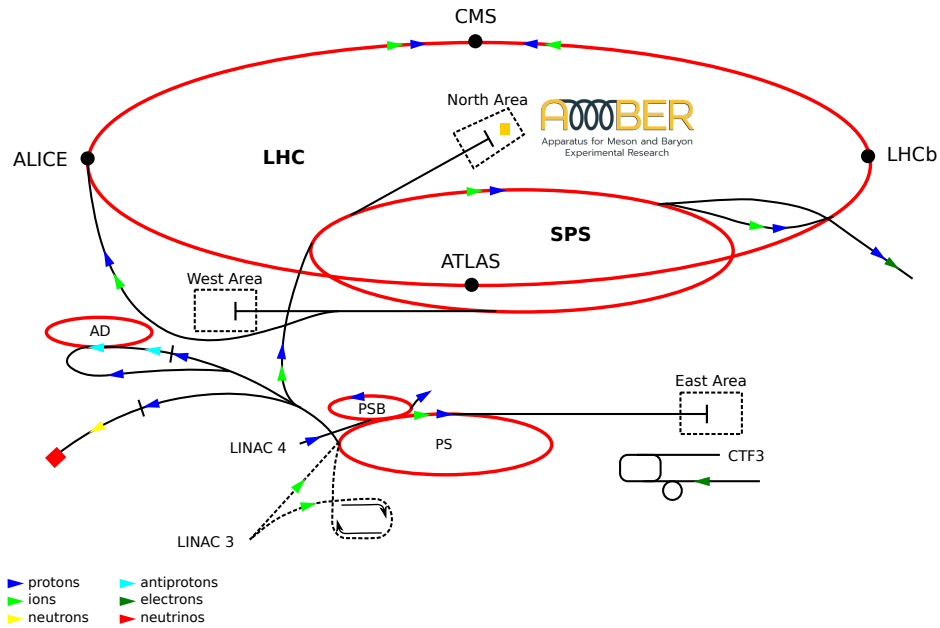


Figure 2.1: Location of the AMBER experiment within the CERN accelerator complex. Figure adapted from [1].

2.2 Experimental Setup for the Proton Radius Measurement

The proton-radius measurement is performed using a 100 GeV muon beam. The beam is delivered to the detector in spills. A spill refers to a single period during which the beam is extracted from the accelerator and delivered to the experiment. For AMBER, a spill is approximately 4.8 s long. Therefore, an intensity of approximately 4×10^6 muons per spill corresponds to an average rate of about 8.3×10^5 muons per second that hit the target during the spill.

The beam is directed onto an active hydrogen target operated at a pressure of 20 bar. The high gas pressure increases the target density and therefore the probability of muon–proton interactions. Since hydrogen provides a clean proton target, it allows the proton kinematics to be reconstructed without additional nuclear effects.

The recoil proton is reconstructed inside the target using a Time Projection Chamber (TPC), while the incoming and scattered muons are measured by dedicated tracking detectors and the downstream spectrometer. An overview of the experimental setup is shown in Figure 2.2.

The direction of the incoming beam defines the longitudinal axis of the experiment. Detector components located before the interaction point with respect to the beam direction are referred to as upstream, whereas components located after the interaction point are called downstream.

The main detector components relevant for the proton-radius measurement are described in the following.

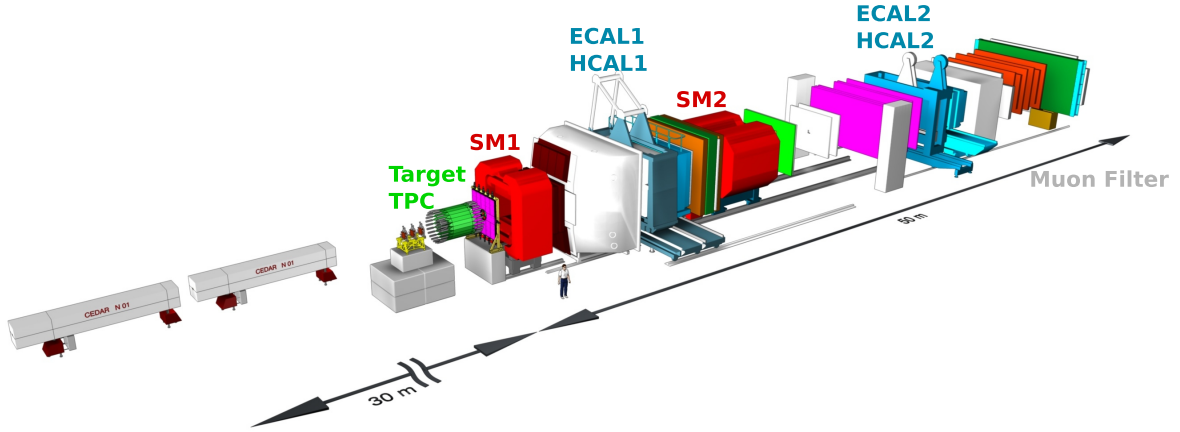


Figure 2.2: Overview of the AMBER setup, with the detector components relevant for the proton-radius measurement labeled. Figure adapted from [XXX].

Unified Tracking Stations Four Unified Tracking Stations (UTS) surround the target region, with two stations located upstream and two downstream of the TPC. Their main purpose is to reconstruct the trajectories of the incoming and scattered muons and thereby determine the small muon scattering angles with high precision.

Each UTS combines Silicon Pixel Detectors (SPD) based on ALPIDE Monolithic Active Pixel Sensors (MAPS) with Scintillating Fiber Hodoscopes (SFH). The ALPIDE sensors provide

the high spatial resolution required for precise track reconstruction. Their small pixel size of approximately $28 \times 28 \mu\text{m}^2$ allows the position of the traversing muon to be measured accurately.

The SFH complement the pixel detectors by providing precise timing information. This is particularly important because the comparatively long integration time of the ALPIDE sensors can result in hits from several beam particles being recorded within the same time window. The time information provided by the SFH allows these ambiguities to be resolved and helps to associate the reconstructed muon tracks with the corresponding event in the TPC.

Time Projection Chamber The TPC serves as an active hydrogen target. It is operated with hydrogen gas at a pressure of 20 bar and is divided into two drift cells, each with a length of 40 cm. In total, the system contains approximately 20 m^3 of hydrogen. Thin carbon-fiber composite windows are used where the muon beam enters and leaves the TPC in order to minimize interactions of the beam particles with detector material.

When a charged recoil proton traverses the hydrogen gas, it ionizes the gas molecules and creates electron-ion pairs along its trajectory. A high voltage of approximately 100 kV generates the electric drift field, causing the liberated electrons to drift towards the readout plane. A Frisch grid is placed in front of the anode to shield it from the drifting electrons until they reach the readout region. This reduces the dependence of the measured signal on the position at which the ionization was produced and improves the energy and time resolution.

The anode is segmented into pads, allowing the position and arrival time of the electrons to be measured and the three-dimensional proton track to be reconstructed.

In addition to the track direction, the energy deposited by the proton provides information about its kinetic energy. For an initial proton at rest, the squared four-momentum transfer is directly related to the recoil-proton kinetic energy. The TPC therefore provides direct access to the recoil-proton kinematics and to the momentum transfer of the scattering process.

To maintain stable detector operation, the hydrogen is continuously recirculated and filtered, keeping impurity levels below approximately 1 ppm.

Spectrometer Magnets Downstream of the target region, the AMBER spectrometer contains two dipole magnets, SM1 and SM2. For the proton-radius measurement, the first magnet, SM1, is switched off, while the second dipole magnet, SM2, is operated. The deflection in SM2, together with downstream tracking detectors such as the GEM stations, allows the momentum of the scattered muon to be reconstructed.

Since elastic muon-proton scattering in the investigated low- Q^2 region is strongly forward-peaked, precise reconstruction of the small scattering angles and the high-momentum outgoing muon is essential.

Electromagnetic and Hadronic Calorimeters The AMBER spectrometer features a two-stage calorimeter system containing two electromagnetic calorimeters (ECAL1 and ECAL2) and

two hadronic calorimeters (HCAL1 and HCAL2). The calorimeters measure the energy of particles through the detection of the energy deposited in active detector material.

Electromagnetic calorimeters are primarily sensitive to electrons and photons, which produce electromagnetic showers in the detector material, whereas hadronic calorimeters are designed to measure strongly interacting particles.

For the proton-radius measurement, ECAL2 is particularly important because it provides sensitivity to photons emitted in radiative muon–proton scattering. Such photon emission modifies the kinematics of the observed final state and therefore has to be taken into account in the determination of radiative corrections. Although ECAL2 did not provide data for the proton-radius run before Long Shutdown 3, future measurements will be important for comparisons with theoretical predictions and the simulation framework developed in this thesis.

Muon Filters Muon identification is provided by the downstream AMBER spectrometer. Dedicated muon-filter systems consist of thick absorber material followed by tracking detectors. Hadrons and electromagnetic shower components are strongly attenuated in the absorber, whereas high-energy muons can penetrate the material. Tracks detected downstream of the absorber can therefore be matched to reconstructed spectrometer tracks to identify the scattered muon.

3 Theoretical Background on Elastic Muon-Proton Scattering

Throughout this chapter, natural units with $\hbar = c = 1$ and the metric convention $g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$ are used.

3.1 From Transition Rates to Scattering Cross Sections

The transition rate of a quantum mechanical process is described by Fermi's golden rule. It relates the probability of a transition between an initial state and a final state to the interaction matrix element and the available phase space:

$$\Gamma_{i \rightarrow f} = 2\pi |\mathcal{M}_{fi}|^2 \rho(E_f), \quad (3.1)$$

where $\mathcal{M}_{fi}$ is the matrix element of the interaction and contains all information about the dynamics of the process. It is calculated from the corresponding Feynman diagrams. The factor $\rho(E_f)$ describes the density of accessible final states and contains only kinematic information. It depends on the masses, energies, and momenta of the particles involved. A larger available phase space increases the number of possible final states and therefore enhances the transition probability. The nonrelativistic matrix element $\mathcal{M}_{fi}$ and the invariant amplitude $\mathcal{M}$ used below play analogous roles, although their state-normalization conventions are different.

For the decay of one initial particle with four-momentum p_1 into n final-state particles, the differential transition rate is given by

$$d\Gamma = \frac{(2\pi)^4}{2E_1} |\mathcal{M}|^2 \delta^{(4)} \left(p_1 - \sum_f p_f \right) d\Phi_n, \quad (3.2)$$

where the Lorentz-invariant phase space element is defined as

$$d\Phi_n = \prod_f \frac{d^3 \mathbf{p}_f}{(2\pi)^3 2E_f}. \quad (3.3)$$

The four-dimensional delta function ensures conservation of energy and momentum. In scattering processes, the relevant observable is not the transition rate but the cross section. It is obtained by normalizing the transition rate to the incident particle flux,

$$d\sigma = \frac{d\Gamma}{\text{flux}}. \quad (3.4)$$

The flux factor accounts for the relative motion of the incoming particles and normalizes the transition rate to the number of possible collisions. For a general $2 \rightarrow n$ scattering process, this results in the Lorentz-invariant expression

$$d\sigma = \frac{(2\pi)^4 \overline{|\mathcal{M}|^2}}{4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}} \delta^{(4)}\left(p_1 + p_2 - \sum_f p_f\right) d\Phi_n. \quad (3.5)$$

For an unpolarized process, the bar denotes an average over the initial-state spins and a sum over the final-state spins.

3.2 Non-Radiative Kinematics

In this thesis, only elastic muon-proton scattering is considered,

$$p(p_1) + \mu(p_2) \rightarrow p(p_3) + \mu(p_4). \quad (3.6)$$

The proton is initially at rest, therefore the incoming four-momenta are given by

$$p_1 = (m_p, \mathbf{0}), \quad p_2 = (E_\mu, \mathbf{p}_\mu), \quad (3.7)$$

where m_p is the proton mass and E_μ is the energy of the incoming muon. The outgoing proton and muon are described by

$$p_3 = (E'_p, \mathbf{p}'_p), \quad p_4 = (E'_\mu, \mathbf{p}'_\mu). \quad (3.8)$$

Energy and momentum conservation require

$$p_1 + p_2 = p_3 + p_4. \quad (3.9)$$

The interaction is mediated by the exchange of a virtual photon with four-momentum

$$q = p_2 - p_4 = p_3 - p_1. \quad (3.10)$$

The invariant momentum transfer is defined as

$$Q^2 = -q^2, \quad (3.11)$$

where the positive sign convention for Q^2 is commonly used in scattering experiments. The value of Q^2 determines the resolution scale of the interaction: larger values of Q^2 correspond to a smaller distance scale probed inside the proton.

The Mandelstam variables provide a convenient description of relativistic scattering processes. Because these variables are Lorentz invariant, they give a compact, frame-independent representation of scattering kinematics and cross sections. For a $2 \rightarrow 2$ process, they are

$$s = (p_1 + p_2)^2 = (p_3 + p_4)^2 = m_p^2 + 2m_p E_\mu + m_\mu^2,$$

$$t = (p_1 - p_3)^2 = (p_2 - p_4)^2 = 2m_p^2 - 2m_p E'_p = -Q^2, \quad (3.12)$$

$$u = (p_1 - p_4)^2 = (p_2 - p_3)^2 = m_p^2 - 2m_p E'_\mu + m_\mu^2.$$

They satisfy

$$s + t + u = 2m_p^2 + 2m_\mu^2.$$

3.3 Differential Cross Section for Elastic Muon-Proton Scattering

The differential cross section for elastic muon-proton scattering can be expressed in terms of the Mandelstam variables and the matrix element of the interaction. The general expression for a two-particle scattering process is given by

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s} \frac{1}{|\mathbf{p}_{1,\text{cm}}|^2} |\overline{\mathcal{M}}|^2,$$

where $|\mathbf{p}_{1,\text{cm}}|$ is the magnitude of the incoming particle momentum in the center-of-mass frame. To express the momentum in terms of the Mandelstam variable s , the Källén function is introduced:

$$\lambda(s, m_\mu^2, m_p^2) = s^2 + m_\mu^4 + m_p^4 - 2s(m_\mu^2 + m_p^2) - 2m_\mu^2 m_p^2.$$

The center-of-mass momentum is related to the Källén function by

$$|\mathbf{p}_{1,\text{cm}}|^2 = \frac{\lambda(s, m_\mu^2, m_p^2)}{4s}.$$

Therefore, the differential cross section can be rewritten as

$$\frac{d\sigma}{dt} = \frac{1}{16\pi\lambda(s, m_\mu^2, m_p^2)} |\overline{\mathcal{M}}|^2.$$

This expression is written in a Lorentz-invariant form and can therefore be evaluated in any reference frame.

However, the experimentally measured quantity is not directly $d\sigma/dt$. The detector measures the kinematic properties of the outgoing muon, in particular its scattering angle and energy. Therefore, the differential cross section is usually expressed as a function of the scattering angle:

$$\frac{d\sigma}{d\Omega} = \frac{d\sigma}{dt} \left| \frac{dt}{d\Omega} \right|. \quad (3.13)$$

Since $t = -Q^2$ by definition, see Equation 3.12, the absolute value of the transformation factor is

$$\left| \frac{dt}{d\Omega} \right| = \frac{dQ^2}{d\Omega}. \quad (3.14)$$

No massless approximation is made for the muon in the following derivation. In the laboratory frame, the magnitudes of the incoming and outgoing muon momenta are

$$p_\mu = \sqrt{E_\mu^2 - m_\mu^2}, \quad p'_\mu = \sqrt{E_\mu'^2 - m_\mu^2}. \quad (3.15)$$

The on-shell condition for the final-state proton, $(p_1 + q)^2 = m_p^2$, gives

$$Q^2 = 2m_p(E_\mu - E'_\mu). \quad (3.16)$$

Independently, evaluating $Q^2 = -(p_2 - p_4)^2$ yields

$$Q^2 = 2(E_\mu E'_\mu - p_\mu p'_\mu \cos \theta_\mu - m_\mu^2), \quad (3.17)$$

where θ_μ is the laboratory scattering angle of the outgoing muon. Equations (3.16) and (3.17) determine E'_μ and Q^2 for a given beam energy and scattering angle. Solving them for the physical outgoing-muon energy gives

$$E'_\mu = \frac{(m_p + E_\mu)(m_p E_\mu + m_\mu^2) + p_\mu^2 \cos \theta_\mu \sqrt{m_p^2 - m_\mu^2 \sin^2 \theta_\mu}}{(m_p + E_\mu)^2 - p_\mu^2 \cos^2 \theta_\mu}. \quad (3.18)$$

For an azimuthally symmetric unpolarized cross section, the azimuthal angle has already been integrated from 0 to 2π , so that

$$d\Omega = 2\pi \sin \theta_\mu d\theta_\mu. \quad (3.19)$$

Using this, the transformation between the two differential cross sections becomes

$$\frac{dQ^2}{d\Omega} = \frac{1}{2\pi \sin \theta_\mu} \left| \frac{dQ^2}{d\theta_\mu} \right|.$$

Differentiating the exact kinematic relations gives

$$\frac{dQ^2}{d\theta_\mu} = \frac{2m_p p_\mu p_\mu'^2 \sin \theta_\mu}{p_\mu' (m_p + E_\mu) - p_\mu E_\mu' \cos \theta_\mu}. \quad (3.20)$$

Consequently, the exact transformation factor is

$$\frac{dQ^2}{d\Omega} = \frac{m_p p_\mu p_\mu'^2}{\pi [p_\mu' (m_p + E_\mu) - p_\mu E_\mu' \cos \theta_\mu]}. \quad (3.21)$$

In the laboratory frame, the Källén function satisfies $\lambda(s, m_\mu^2, m_p^2) = 4m_p^2 p_\mu^2$. Inserting this relation and Equation 3.21 into the invariant cross section gives

$$\frac{d\sigma}{d\Omega} = \frac{p_\mu'^2}{64\pi^2 m_p p_\mu [p_\mu' (m_p + E_\mu) - p_\mu E_\mu' \cos \theta_\mu]} |\overline{\mathcal{M}}|^2. \quad (3.22)$$

This expression contains the exact two-body phase-space and recoil factors for a finite muon mass. The remaining task is to determine the matrix element $\mathcal{M}$ describing the electromagnetic interaction.

In the ultra-relativistic limit $E_\mu \gg m_\mu$. It follows that $p_\mu \approx E_\mu$ and the outgoing muon momentum can be approximated by $p'_\mu \approx E'_\mu$. In this limit

$$\frac{dQ^2}{d\theta_\mu} = \frac{2E_\mu^2 \sin \theta_\mu}{[1 + (2E_\mu/m_p) \sin^2(\theta_\mu/2)]^2}. \quad (3.23)$$

and the differential cross section simplifies to

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 m_p^2 [1 + (2E_\mu/m_p) \sin^2(\theta_\mu/2)]^2} |\mathcal{M}|^2. \quad (3.24)$$

3.4 Born Approximation

The starting point for calculating the scattering amplitude is the interaction part of the QED Lagrangian. At leading order in quantum electrodynamics (Born approximation), the interaction is described by the exchange of a single virtual photon.

$$\mathcal{L}_{\text{int}} = -e Q_f \bar{\psi} \gamma^\mu \psi A_\mu,$$

where ψ is the Dirac spinor field representing the charged fermion, A_μ is the electromagnetic four-potential, γ^μ are the Dirac gamma matrices, $e > 0$ is the elementary charge, and Q_f is the fermion charge in units of e . The interaction Lagrangian can therefore be written in terms of the electromagnetic current $j_{\text{em}}^\mu = Q_f \bar{\psi} \gamma^\mu \psi$ as

$$\mathcal{L}_{\text{int}} = -e j_{\text{em}}^\mu A_\mu.$$

This expression directly shows that the electromagnetic interaction arises from the coupling between the electromagnetic current and the electromagnetic field. In a scattering process, this coupling is represented diagrammatically by a vertex. The vertex therefore describes the interaction between a charged fermion and a photon. For a point-like fermion such as the muon, the electromagnetic vertex follows directly from the QED interaction Lagrangian. Apart from the charge factor Q_μ , the corresponding Feynman rule is

$$-ie\gamma^\mu,$$

and, written in terms of the incoming and outgoing muon spinors, the vertex factor is

$$Q_\mu \bar{u}(p_4) (-ie\gamma^\mu) u(p_2),$$

where $u(p_2)$ and $\bar{u}(p_4)$ are the Dirac spinors for the incoming and outgoing muon, respectively.

For a composite particle like the proton, the electromagnetic vertex is modified by the internal structure of the particle. This modification is described by the electromagnetic form

factors, which encode information about the spatial distribution of charge and magnetization within the proton. The vertex factor for the proton can be expressed as

$$\bar{u}(p_3)(-ie\Gamma^\nu(Q^2))u(p_1),$$

where $u(p_1)$ and $\bar{u}(p_3)$ are the Dirac spinors for the incoming and outgoing proton. In the spacelike region, the form factors are written as functions of the positive variable $Q^2 = -q^2$; that is, $F_i(Q^2) \equiv F_i(q^2 = -Q^2)$. The electromagnetic vertex function is

$$\Gamma^\nu(Q^2) = F_1(Q^2)\gamma^\nu + \frac{i\sigma^{\nu\rho}q_\rho}{2m_p}F_2(Q^2),$$

where $\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho]$ is the antisymmetric tensor constructed from the gamma matrices, and m_p is the mass of the proton. The Q^2 dependence of the form factors is shown in Figure 3.1.

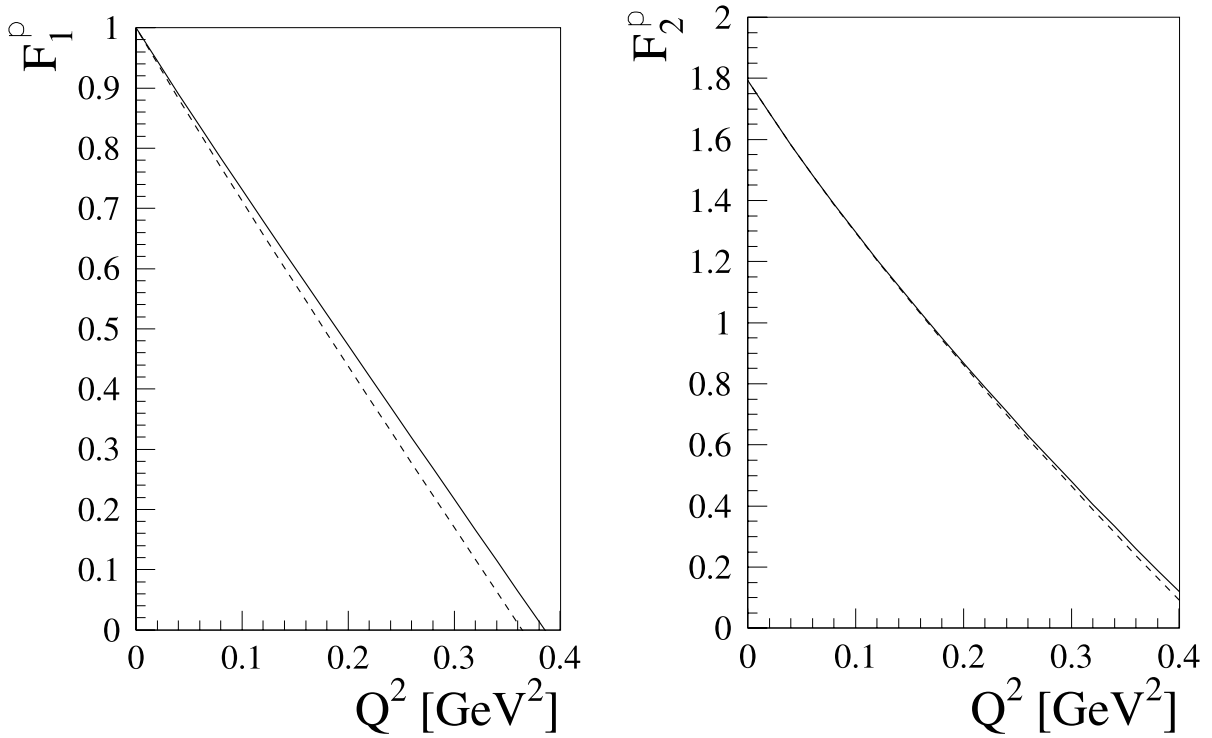


Figure 3.1: The Dirac (F_1) and Pauli (F_2) form factors of the proton in relativistic chiral perturbation theory at $\mathcal{O}(q^4)$. Comparison of two different calculation approaches: EOMS (solid lines) and infrared regularization (dashed lines). Figures taken from Thomas Fuchs 2004.

At zero momentum transfer, the Dirac and Pauli form factors are normalized as

$$F_1(0) = 1, \quad F_2(0) = \mu_p - 1,$$

where μ_p is the nuclear magneton. The two form factors together describe the response of the proton's charge and magnetization structure to the electromagnetic probe. A more direct interpretation in terms of electric and magnetic distributions is provided by the Sachs form factors.

The Sachs form factors are related to the Dirac and Pauli form factors by

$$G_E(Q^2) = F_1(Q^2) - \frac{Q^2}{4m_p^2} F_2(Q^2)$$

and

$$G_M(Q^2) = F_1(Q^2) + F_2(Q^2).$$

The Q^2 dependence of the Sachs form factors is shown in Figure 3.2. The exchanged photon

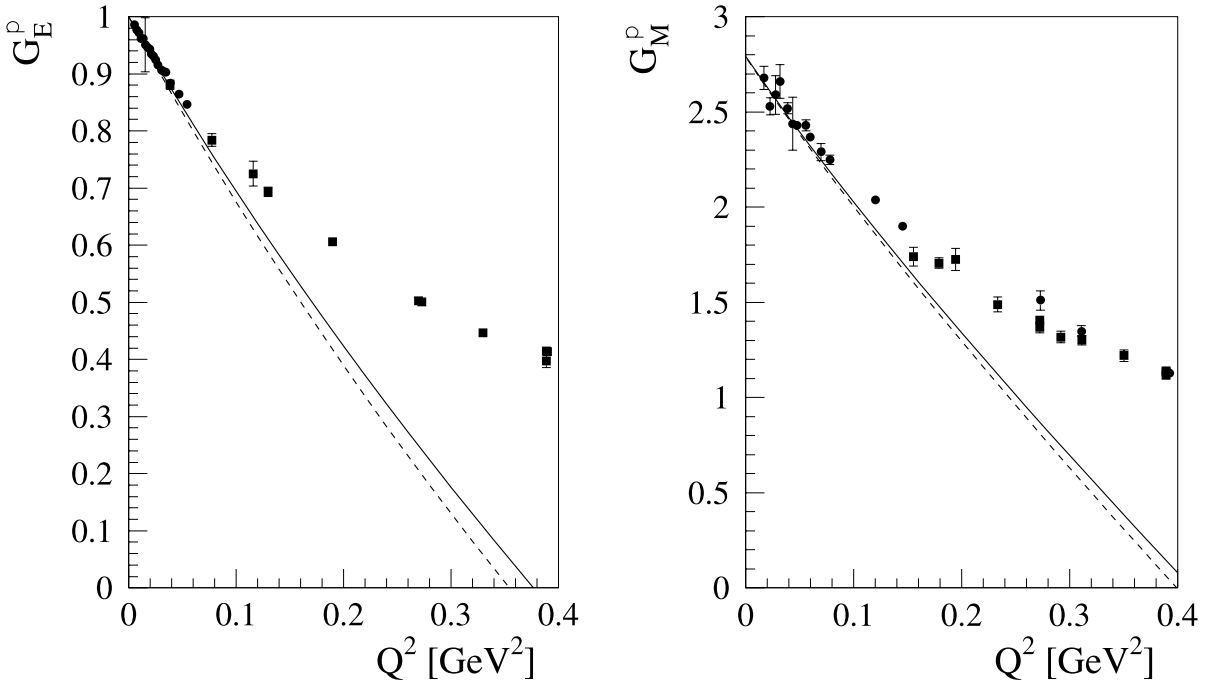


Figure 3.2: The Sachs form factors of the nucleon at $\mathcal{O}(q^4)$. Comparison of two different calculation approaches: EOMS (solid lines) and infrared regularization (dashed lines). Figures taken from Thomas Fuchs 2004 [XXX].

connects the muon and proton vertices. Since this photon is virtual, it is described by a photon propagator rather than by an external photon wave function. In Feynman gauge, the propagator is

$$\frac{-ig_{\mu\nu}}{q^2},$$

where $g_{\mu\nu}$ is the metric tensor. The muon and proton vertices carry the Lorentz indices μ and ν , respectively. These indices are contracted through the virtual-photon propagator,

whose metric tensor $g_{\mu\nu}$ connects the leptonic and hadronic currents according to the Einstein summation convention.

Written with both Lorentz indices explicitly, the matrix element is

$$\mathcal{M} = [Q_\mu \bar{u}(p_4)(-ie\gamma^\mu)u(p_2)] \frac{-ig_{\mu\nu}}{q^2} [\bar{u}(p_3)(-ie\Gamma^\nu(Q^2))u(p_1)].$$

The contraction with the metric tensor identifies the two Lorentz indices. In particular,

$$g_{\mu\nu}\Gamma^\nu(Q^2) = \Gamma_\mu(Q^2),$$

where the index is lowered according to the metric. Thus, the propagator effectively connects the Lorentz index of the leptonic current to that of the hadronic current. The matrix element can therefore be written as

$$\mathcal{M} = Q_\mu \frac{e^2}{q^2} \bar{u}(p_4)\gamma^\mu u(p_2) \bar{u}(p_3)\Gamma_\mu(Q^2)u(p_1).$$

This expression is the starting point for calculating the cross section. However, the matrix element itself is not yet the observable quantity. The differential cross section is proportional to the squared modulus of the matrix element, summed over the unobserved final-state spins and averaged over the initial-state spins.

For unpolarized scattering, the spin-averaged squared matrix element is defined as

$$\overline{|\mathcal{M}|^2} = \frac{1}{4} \sum_{s_1, s_2, s_3, s_4} |\mathcal{M}|^2,$$

where the factor $1/4$ averages over the two possible initial spin states of the muon and the two possible initial spin states of the proton.

Writing the spin labels explicitly, the matrix element is

$$\mathcal{M} = Q_\mu \frac{e^2}{q^2} [\bar{u}(p_4, s_4)\gamma^\mu u(p_2, s_2)] [\bar{u}(p_3, s_3)\Gamma_\mu(Q^2)u(p_1, s_1)].$$

Its complex conjugate is

$$\mathcal{M}^* = Q_\mu \frac{e^2}{q^2} [\bar{u}(p_2, s_2)\gamma^\nu u(p_4, s_4)] [\bar{u}(p_1, s_1)\bar{\Gamma}_\nu(Q^2)u(p_3, s_3)],$$

where

$$\bar{\Gamma}_\nu = \gamma^0 \Gamma_\nu^\dagger \gamma^0.$$

Therefore,

$$|\mathcal{M}|^2 = \frac{e^4}{(q^2)^2} [\bar{u}(p_4)\gamma^\mu u(p_2)] [\bar{u}(p_2)\gamma^\nu u(p_4)] [\bar{u}(p_3)\Gamma_\mu u(p_1)] [\bar{u}(p_1)\bar{\Gamma}_\nu u(p_3)].$$

To evaluate the spin sums, the completeness relation for Dirac spinors is used:

$$\sum_s u(p, s)\bar{u}(p, s) = \not{p} + m,$$

where

$$\not{p} = \gamma^\mu p_\mu.$$

This relation allows the products of spinors to be converted into traces over products of gamma matrices. The spin-averaged squared matrix element can consequently be written as

$$|\overline{\mathcal{M}}|^2 = \frac{e^4}{4(q^2)^2} L^{\mu\nu} H_{\mu\nu},$$

where the leptonic tensor is

$$L^{\mu\nu} = \text{Tr} [(\not{p}_4 + m_\mu) \gamma^\mu (\not{p}_2 + m_\mu) \gamma^\nu],$$

and the hadronic tensor is

$$H_{\mu\nu} = \text{Tr} [(\not{p}_3 + m_p) \Gamma_\mu(Q^2) (\not{p}_1 + m_p) \bar{\Gamma}_\nu(Q^2)].$$

The factorization into a leptonic and a hadronic tensor separates the two physical contributions. The leptonic tensor describes the coupling of the virtual photon to the muon, while the hadronic tensor contains the proton electromagnetic structure through the form factors.

For the point-like muon, the leptonic tensor can be evaluated using the standard gamma-matrix trace identities. This gives

$$L^{\mu\nu} = 4 \left[p_4^\mu p_2^\nu + p_4^\nu p_2^\mu - g^{\mu\nu} (p_2 \cdot p_4 - m_\mu^2) \right]. \quad (3.25)$$

The leptonic tensor obeys current conservation, $q_\mu L^{\mu\nu} = q_\nu L^{\mu\nu} = 0$.

To evaluate the hadronic tensor, it is convenient to introduce

$$\tau = \frac{Q^2}{4m_p^2} \quad (3.26)$$

and the sum of the initial and final proton momenta,

$$P^\mu = p_1^\mu + p_3^\mu. \quad (3.27)$$

The on-shell conditions imply $P \cdot q = 0$ and $P^2 = 4m_p^2 + Q^2$. Using the Sachs form factors, the hadronic trace can then be written in the compact form

$$H_{\mu\nu} = 2 \frac{G_E^2(Q^2) + \tau G_M^2(Q^2)}{1 + \tau} P_\mu P_\nu - 2G_M^2(Q^2) (q_\mu q_\nu + Q^2 g_{\mu\nu}). \quad (3.28)$$

The terms containing $q_\mu q_\nu$ do not contribute to the contraction with the conserved leptonic tensor. Defining

$$A(Q^2) = \frac{G_E^2(Q^2) + \tau G_M^2(Q^2)}{1 + \tau}, \quad (3.29)$$

the contraction of Equation 3.25 with Equation 3.28 yields the exact spin-averaged squared Born matrix element

$$\begin{aligned} |\overline{\mathcal{M}_{\text{Born}}}|^2 = \frac{e^4}{Q^4} \Big\{ & A(Q^2) \left[(s-u)^2 - Q^2 (4m_p^2 + Q^2) \right] \\ & + 2G_M^2(Q^2) Q^2 (Q^2 - 2m_\mu^2) \Big\}. \end{aligned} \quad (3.30)$$

Inserting Equation 3.30 into Equation 3.24 gives the exact laboratory-frame Born cross section,

$$\begin{aligned} \frac{d\sigma_{\text{Born}}}{d\Omega} = \frac{1}{64\pi^2 m_p p_\mu} \frac{p'_\mu{}^2}{p'_\mu(m_p + E_\mu) - p_\mu E'_\mu \cos \theta_\mu} \frac{e^4}{Q^4} \\ \times \Big\{ A(Q^2) \left[(s-u)^2 - Q^2 (4m_p^2 + Q^2) \right] + 2G_M^2(Q^2) Q^2 (Q^2 - 2m_\mu^2) \Big\}. \end{aligned} \quad (3.31)$$

In the ultrarelativistic limit, the Leptonic tensor simplifies to

$$L^{\mu\nu} = 4 (p_4^\mu p_2^\nu + p_4^\nu p_2^\mu - g^{\mu\nu} p_4 \cdot p_2).$$

and the hadronic tensor reduces to !!!!!check this!!!!!!

$$H_{\mu\nu} = 2 \frac{G_E^2(Q^2) + \tau G_M^2(Q^2)}{1 + \tau} P_\mu P_\nu.$$

Evaluating the leptonic and hadronic tensors and performing the contraction $L^{\mu\nu} H_{\mu\nu}$ then leads to the Rosenbluth form of the elastic Born cross section.

$$\frac{d\sigma_{\text{Born}}}{d\Omega} = \frac{d\sigma_{\text{Mott}}}{d\Omega} \frac{\tau}{\epsilon(1 + \tau)} \left(G_M^2(Q^2) + \frac{\epsilon}{\tau} G_E^2(Q^2) \right), \quad (3.32)$$

where $\frac{d\sigma_{\text{Mott}}}{d\Omega}$ is the Mott cross section which describes the elastic scattering of a charged lepton from a structureless charged target. In the ultrarelativistic limit, it is given by

$$\frac{d\sigma_{\text{Mott}}}{d\Omega} = \frac{\alpha^2 \cos^2(\theta/2)}{4E^2 \sin^4(\theta/2)}. \quad (3.33)$$

, where the the so-called virtual photon polarization parameter ϵ is given by

$$\epsilon = \left(1 + 2(1 + \tau) \tan^2 \frac{\theta}{2} \right)^{-1}. \quad (3.34)$$

In the low- Q^2 region relevant for proton radius measurements, the Sachs form factors can be approximated by the empirical dipole parametrization

$$G_D(Q^2) = \left(1 + \frac{Q^2}{\Lambda^2} \right)^{-2}, \quad (3.35)$$

where the dipole mass parameter is

$$\Lambda^2 = 0.71 \text{ GeV}^2. \quad (3.36)$$

The electric and magnetic form factors are then expressed as

$$G_E(Q^2) = G_D(Q^2) = \left(1 + \frac{Q^2}{0.71 \text{ GeV}^2}\right)^{-2},^1 \quad (3.37)$$

and

$$G_M(Q^2) = \mu_p G_D(Q^2) = \mu_p \left(1 + \frac{Q^2}{0.71 \text{ GeV}^2}\right)^{-2}, \quad (3.38)$$

where $\mu_p = 2.792847$ is the dimensionless proton magnetic moment in units of the nuclear magneton μ_N . It defines the normalization of the magnetic form factor, such that $G_M(0) = \mu_p$.

3.5 Radiative Corrections

This thesis considers radiative corrections through next-to-leading order (NLO) in QED. The Born amplitude is of order e^2 and gives a cross section of order α^2 . The NLO contribution to the cross section is of order α^3 and receives contributions from both virtual one-loop amplitudes and real-photon emission. The corresponding Feynman diagrams are shown in Figure 3.3.

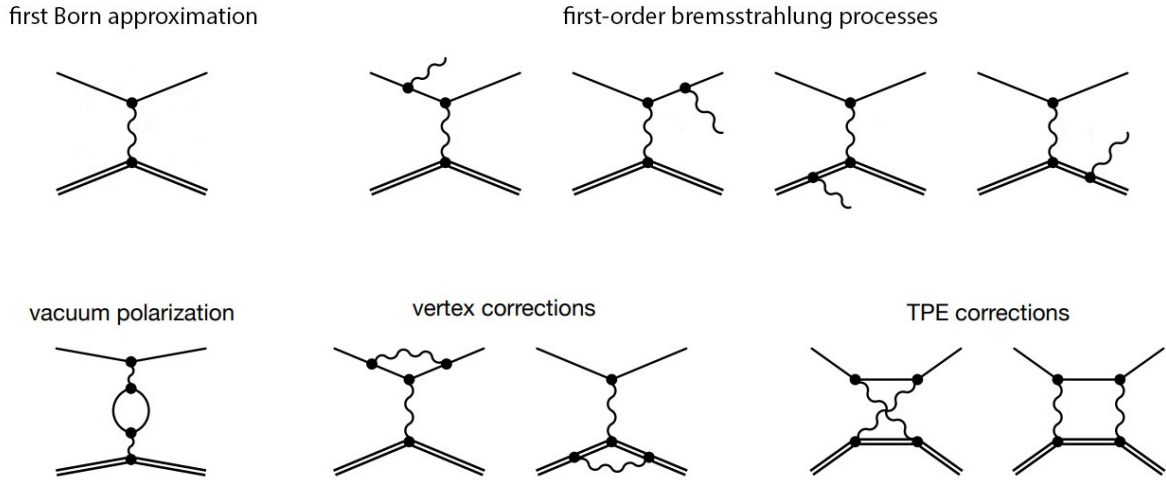


Figure 3.3: Leading-order and next-to-leading-order contributions to elastic muon-proton scattering. Figure provided by M. Depour.

¹For a point-like proton, the form factor would be $G_E(Q^2) = 1$ for all Q^2 .

Virtual corrections. Virtual corrections arise from one-loop diagrams and include self-energy, vertex, vacuum-polarization, and, where included in the chosen correction class, two-photon-exchange contributions. They modify the Born amplitude through the interference term between the one-loop and Born amplitudes. Ultraviolet divergences are removed by renormalization, whereas infrared divergences cancel only after the corresponding real-emission contribution is included.

Real photon emission. The real contribution is the bremsstrahlung process $p\mu \rightarrow p\mu\gamma$, in which an additional photon is emitted from a charged external particle. Its phase space must be combined with the virtual contribution using the same definition of the measured observable. The resulting sum is infrared finite. Experimentally, its numerical value also depends on acceptance requirements and photon-energy cuts.

4 Simulation with McMule

5 Results

6 Discussion

7 Conclusion and Outlook

Diese Arbeit war ganz toll.

Bibliography

- [1] Oleg Denisov. *Super Proton Synchrotron*. 0. URL: https://de.wikipedia.org/wiki/Super_Proton_Synchrotron (visited on 07/16/2026).